

# Geometric Rigidity and Axiomatic Inconsistency: A Synthesis of Lorentzian Dynamics and Partial Information Decomposition

## 1. Introduction: The Search for Invariant Structures in Complex Systems

The theoretical landscape of late 2025 is defined by a profound convergence of two previously disparate domains: the differential geometry of relativistic spacetimes and the information-theoretic decomposition of multivariate dependencies. At the heart of this convergence lies a shared epistemological crisis and resolution—the tension between local flexibility and global consistency. In the geometric realm, this is encapsulated by the **Lorentzian Lichnerowicz Conjecture**, a proposition regarding the rigidity of causal structures under conformal transformations. In the informational realm, it manifests in the **Partial Information Decomposition (PID)** framework, particularly the **Novel Inconsistency Results** published by Michael Wibral and colleagues in December 2025.

For decades, the "Shannon bit"—the fundamental unit of uncertainty resolution—was treated as a monolithic scalar quantity, sufficient for characterizing point-to-point communication channels. However, the study of complex adaptive systems, such as the neural connectome or the fabric of quantum spacetime, revealed that information in a multivariate setting possesses an internal structure that Shannon's original formalism obscures. The decomposition of this information into **redundant**, **unique**, and **synergistic** atoms requires a rigorous axiomatic foundation. The seminal work of Wibral et al. in late 2025 has demonstrated that such a foundation is mathematically fraught with contradictions; specifically, that the desirable properties of non-negativity, chain-rule additivity, and transformation invariance cannot coexist within the PID framework.<sup>1</sup>

This report posits that the "fit" of Lorentzian manifolds and the Lichnerowicz Conjecture within this informational context is not merely analogical but structural. Both fields grapple with the concept of **rigidity**. The Lichnerowicz Conjecture establishes that Lorentzian manifolds (the substrates of causal information flow) are geometrically rigid; they do not admit essential conformal symmetries unless they are trivial (conformally flat). Similarly, the Wibral Inconsistency Theorems establish that the lattice of information atoms is axiomatically rigid; one cannot smoothly transform between different views of information (via the chain rule or invertible transformations) without "tearing" the logical fabric (violating

non-negativity).

This document provides an exhaustive analysis of these interrelations. It dissects the mechanics of the redundancy lattice in 3-way hierarchical PID, explores the ontological status of the Shannon bit in light of the 2025 inconsistency results, and maps these informational dynamics onto the causal substrate defined by Lorentzian geometry.

## 2. The Geometric Substrate: Lorentzian Manifolds and Causal Rigidity

To understand the constraints on information flow, one must first understand the stage upon which it propagates. In physical theories, this stage is a **Lorentzian manifold**, the mathematical model of spacetime in General Relativity. The geometric properties of this manifold dictate the "allowable" pathways for information, defining the causal structure that PID attempts to quantify.

### 2.1 The Lorentzian Signature and Causal Structure

A Lorentzian manifold  $(M, g)$  differs from a Riemannian manifold in its metric signature. While a Riemannian metric is positive-definite  $(+, +, \dots, +)$ , a Lorentzian metric has a signature of  $(-, +, \dots, +)$ .<sup>3</sup> This single sign difference generates the **light cone structure** at every point  $p \in M$ , dividing the tangent space  $T_p M$  into three distinct causal regions.

This tripartite division is the geometric precursor to the tripartite decomposition of information (Redundancy, Uniqueness, Synergy).

- **Timelike Vectors ( $g(v, v) < 0$ ):** These represent the trajectories of massive observers. In information terms, this is the "channel" through which distinct sources persist and transmit data.
- **Null Vectors ( $g(v, v) = 0$ ):** These define the boundary of the light cone. They represent the absolute speed limit of information transmission. The causal past of an event—the set of all points from which information could have reached  $T$ —is defined strictly by these null geodesics.
- **Spacelike Vectors ( $g(v, v) > 0$ ):** These lie outside the light cone. Events separated by spacelike intervals are causally disconnected; neither can inform the other.

The "Causal Structure" of a spacetime is the partial ordering defined by these relations. If we discretize the manifold, we obtain a **Causal Set**, a locally finite partially ordered set (poset) that retains the logical structure of the spacetime.<sup>4</sup> It is here that the connection to PID begins to materialize: PID is essentially a measure theory defined over a lattice (a specific type of poset), while Lorentzian geometry is the study of the causal poset itself.

### 2.2 The Conformal Group and Information Invariance

Information theory often seeks quantities that are invariant under invertible transformations (relabeling). In geometry, the analogue is Conformal Invariance. A conformal transformation is a diffeomorphism  $\phi: M \rightarrow M$  that preserves the metric up to a scale factor:

$$\phi^* g = e^{2\sigma} g$$

Critically, conformal transformations preserve the causal structure. They map light cones to light cones, preserving the distinction between past, future, and elsewhere. They do not preserve distances (quantities), but they preserve the "possibility" of information exchange. The **Conformal Group** of a manifold, denoted  $\text{Conf}(M, g)$ , is the group of all such transformations. A central question in differential geometry has been: *How large is this group?* If the group is "essential" (non-compact, or containing transformations that cannot be reduced to isometries), it implies the spacetime has a vast amount of symmetry.

## 2.3 The Lorentzian Lichnerowicz Conjecture

The **Lorentzian Lichnerowicz Conjecture** addresses the relationship between this symmetry and the shape of the universe. It serves as a "rigidity theorem."

**The Conjecture:** *If a compact Lorentzian manifold  $(M, g)$  admits an essential group of conformal transformations, then  $(M, g)$  must be conformally flat.*<sup>5</sup>

Being "conformally flat" means the manifold is locally indistinguishable from Minkowski space (flat, empty spacetime). It implies that the curvature tensor (specifically the Weyl tensor) vanishes.

### 2.3.1 Status of the Proof (As of December 2025)

Research in the years leading up to 2025 has solidified the proof of this conjecture for various classes of manifolds, marking a triumph of geometric rigidity theory.

- **Riemannian Case:** The conjecture was proven for Riemannian manifolds by Ferrand and Obata in the 20th century.
- **Lorentzian Progress (2021-2025):** The problem is significantly harder in the Lorentzian case due to the non-compactness of the orthogonal group  $O(1, n)$ . However, works referenced in the 2025 literature<sup>7</sup> by authors such as Melnick, Pecastaing, and Frances have established the validity of the conjecture for:
  - Compact, simply connected, real-analytic Lorentzian manifolds.<sup>7</sup>
  - Three-dimensional real-analytic manifolds.<sup>5</sup>
  - Spatially compact spacetimes with specific causal properties.

### 2.3.2 Implications for Information Theory

The proof of the Lichnerowicz Conjecture implies that **complex causal structures are rigid**.

You cannot have a universe with complex gravity (curvature/interaction) that also allows you to arbitrarily rescale space and time (essential conformal symmetry) without changing the physics.

This mirrors the findings in Wibral's 2025 PID work: You cannot have a complex information structure (Synergy/Redundancy) that allows for arbitrary axiomatic flexibility (Chain Rules/Invariance) without breaking the logical consistency. **Symmetry breaking is the price of complexity.** In geometry, breaking the conformal symmetry allows for curvature (gravity). In information theory, breaking the chain rule or invariance allows for the emergence of distinct semantic atoms (synergy).

### 3. The Crisis of Partial Information Decomposition: Wibral's 2025 Inconsistency Framework

While geometers were proving the rigidity of spacetime, information theorists were discovering the rigidity of logic. The publication of "Novel Inconsistency Results for Partial Information Decomposition" in December 2025 by Matthias, Makkeh, Wibral, and Gutknecht<sup>1</sup> fundamentally altered the understanding of multivariate information.

#### 3.1 The Failure of the Williams-Beer Axioms

The field of PID was initiated by Williams and Beer, who proposed that the total Mutual Information  $I(S; T)$  could be decomposed into a sum of non-negative terms defined over a **Redundancy Lattice**. They proposed a set of axioms to define a measure of redundancy  $I_{\cap}$ .

However, counterexamples quickly emerged showing that the original measures calculated "expected" quantities counter-intuitively. This led to a "zoo" of proposed measures ( $I_{\min}$ ,  $I_{\text{MMI}}$ ,  $I_{\text{BROJA}}$ , etc.), each satisfying some axioms but violating others. The 2025 Wibral paper puts an end to this search for the "perfect" measure by proving that **no such measure can exist** given the standard desiderata of information theory.

#### 3.2 The Inconsistency Trilemma

The core contribution of the 2025 work is the formal proof that three cornerstone properties are mutually incompatible. This "Inconsistency Trilemma" forces researchers to choose which property to sacrifice based on the specific operational context (e.g., neuroscience vs. cryptography).

The three incompatible properties are:

1. Non-Negativity ( $\pi \geq 0$ ):

The requirement that all partial information atoms (redundancy, unique, synergy) must be non-negative. It is ontologically difficult to interpret "negative information" or "negative

synergy."

- *Conflict:* Wibral shows that enforcing the chain rule often forces certain synergistic terms to become negative to balance the equation.

## 2. The Target Chain Rule:

In Shannon theory,  $I(S; T_1, T_2) = I(S; T_1) + I(S; T_2 | T_1)$ .

We expect PID atoms to follow a similar additive logic. If the target  $T$  is split into two parts, the information about the whole should be the sum of the information about the parts (conditioned appropriately).

- *Conflict:* The lattice structure of PID, derived from mereology, creates "overlapping" atoms that do not neatly sum up. Forcing this additivity constrains the measure so tightly that it violates non-negativity.

## 3. Invariance Under Invertible Transformations:

If we apply a reversible code to the sources (e.g., swapping labels 0 and 1, or applying a rotation in the state space), the amount of information should not change.  $\pi(f(S); T) = \pi(S; T)$ .

- *Conflict:* The 2025 paper makes explicit that this "innocent" assumption is actually a strong geometric constraint. When combined with the lattice structure, it restricts the solution space to a point where the other two axioms cannot be met.<sup>2</sup>

## 3.3 Strengthening the Rauh Result

Previously, Rauh et al. had shown an inconsistency involving the "Williams and Beer axioms" and the identity property. Wibral's 2025 work strengthens this by removing the need for the **source monotonicity** condition in the proof.<sup>2</sup> This is significant because source monotonicity was often seen as the "weak link" or the most dispensable axiom. By showing the contradiction persists even without it, Wibral et al. demonstrate that the inconsistency is deep within the structure of **Partial Information** itself, not just an artifact of demanding monotonicity.

## 4. 3-Way Hierarchical PID and the Lattice of Redundancy

The user's query specifically highlights "3-way hierarchical PID." The inconsistency results are most potent and complex in this regime.

### 4.1 From Bivariate to Multivariate Lattices

In a 2-way system (Sources  $S_1, S_2$ , Target  $T$ ), the Redundancy Lattice  $\mathcal{A}$  has only 4 elements (atoms). The Williams-Beer lattice is defined by the partial order of "information specificity."

$$\text{Synergy}_{\{1\}\{2\}}$$

$\nearrow \quad \nwarrow$

$\text{Unique } \{1\} \quad \text{Unique } \{2\}$

$\nwarrow \quad \nearrow$

$\text{Redundancy } \{1, 2\}$

However, in a **3-way system** ( $S_1, S_2, S_3$ ), the combinatorial explosion occurs. The lattice is constructed from the set of all **antichains** of the power set of sources (excluding the empty set). An antichain represents a set of sources that individually provide the *same* information.

For  $n=3$ , the number of atoms jumps to 18 (in the standard Williams-Beer lattice). This hierarchy includes complex terms like:

- **Conditioned Redundancy:** Information shared by  $S_1$  and  $S_2$ , but only when  $S_3$  is known.
- **Graded Synergy:** Information available from  $\{1, 2, 3\}$  jointly, but not from any pair.

## 4.2 The "Degree of Redundancy" $r(\alpha)$

Wibral's 2025 paper introduces refined notation to handle this complexity. They define the **Degree of Redundancy**  $r(\alpha)$  for an atom  $\alpha$ .<sup>1</sup>

- **Synergistic Atoms ( $r(\alpha) = 0$ ):** These atoms contribute to the mutual information  $I(S_{\text{joint}}; T)$  but are "invisible" to the individual marginal mutual informations  $I(S_i; T)$ . In the mereological expansion, they appear with a negative sign in certain decompositions if one tries to force additivity, highlighting the conflict with non-negativity.
- **Redundant Atoms ( $r(\alpha) \geq 2$ ):** These are the "genuinely redundant" terms. They appear in multiple marginal terms. The "over-counting" of these atoms in the sum of marginals is what the redundancy measure seeks to correct.

## 4.3 The "Fit" with Shannon Bits

The concept of the "Shannon bit" undergoes an ontological shift in this framework.

- **Classical View:** A bit is a unit of volume in probability space. It is fungible. A bit from  $S_1$  is equivalent to a bit from  $S_2$  if they reduce the same amount of uncertainty.
- **Hierarchical View:** A "bit" is a **tensor** with position in the redundancy lattice. A "Redundant Bit" is structurally different from a "Unique Bit."
- **The Inconsistency Implication:** We cannot simply define a "Shannon Bit of Synergy." Because of the inconsistency theorem, the magnitude of this bit depends on whether we prioritize the Chain Rule or Non-negativity. The "Shannon bit" ceases to be an absolute invariant in multivariate systems; it becomes frame-dependent (axiom-dependent).

## 5. Synthesis: The Geometric-Informational Isomorphism

Having detailed the geometric and informational sides separately, we now integrate them to answer the core of the user's request: *Where exactly do they fit?*

The connection is found in the shared mathematical structures of **Order Theory** (Lattices/Causal Sets) and **Invariance Groups**.

### 5.1 The Causal Emergence Bridge

The most direct application where these fields meet is the theory of **Causal Emergence** (often associated with Erik Hoel, and referenced in the context of Wibral's work on neural goal functions<sup>10</sup>).

#### The Argument:

1. **Micro-Level:** The brain consists of neurons interacting via physical signals. This interaction is constrained by the **Lorentzian manifold** of spacetime (signals take time to travel, defined by light cones).
2. **Macro-Level:** We group neurons into "modules" or "regions." We treat these as information sources ( $S_1, S_2, \dots$ ).
3. **Decomposition:** We apply 3-way PID to these sources to see if the macro-state holds **Synergistic Information** about the future that is not present in the micro-parts.

#### The "Fit":

The Lorentzian geometry defines the possibility of synergy. For  $S_1, S_2, S_3$  to exhibit synergy about  $T$ , they must lie in the causal past of  $T$  (within the past light cone). However, they must typically be spacelike separated from each other (outside each other's light cones) at the time of encoding, otherwise, they would share information (Redundancy) through direct causation.

Thus, **Synergy requires a specific Lorentzian configuration:** Spacelike separation of sources converging into the Timelike future of the target. This is the "Collider" geometry in causal graphs. The **Lichnerowicz Conjecture** ensures that this causal structure is rigid—one cannot simply "transform away" the separation between sources to make them redundant without fundamentally changing the spacetime (flattening it).

### 5.2 Spectral PID and Lorentzian Functions

A secondary, but distinct, "fit" appears in the signal processing methodology used by Wibral's group.<sup>12</sup>

- **The Methodology:** To analyze neural data (EEG/MEG), researchers compute the Power Spectral Density (PSD).



- **The Function:** The PSD of neural populations is often modeled as a sum of Lorentzian functions (Cauchy distributions), which describe the resonance of damped oscillators (the neural circuits).

$$L(f) = \frac{A}{w^2 + (f - f_0)^2}$$

- **The PID Application:** Wibral applies PID to the *parameters* of these Lorentzian fits (or to the information rates derived from them). This allows for a **Spectrally Resolved PID**.<sup>15</sup>
- **The Connection:** While this uses the Lorentzian *function* rather than the *manifold*, the underlying mathematics of the resonance is derived from the **Green's functions** of the wave equation on the Lorentzian manifold of the neural field.<sup>16</sup> The "Propagator" on the manifold defines the spectral shape of the signal.

## 5.3 Mereology as the Lattice of Spacetime

The **Mereological approach**<sup>17</sup> adopted by Wibral in 2025 provides the final structural link.

- **Spacetime:** In approaches like Causal Set Theory (Quantum Gravity), spacetime is fundamentally a poset (partially ordered set).  $A \prec B$  means  $A$  can influence  $B$ .
- **Information:** In Mereological PID, information is a poset.  $A \preceq B$  means the information in  $A$  is part of the information in  $B$ .

Wibral's inconsistency results can be read as a statement about the **topology of the information lattice**. Just as the topology of a Lorentzian manifold can prevent certain global constructions (e.g., global time coordinates in Gödel spacetimes), the topology of the redundancy lattice prevents the global construction of a signed measure (non-negative PID) that satisfies the Chain Rule.

## 6. Applications and Future Outlook (2025 and Beyond)

The theoretical hardening of PID in 2025 has led to new experimental directions, particularly in Wibral's primary field of neuroscience.

### 6.1 Infomorphic Networks

Understanding that a "perfect" decomposition is impossible, Wibral's group has pivoted to **Infomorphic Networks**.<sup>18</sup> These are artificial neural networks where individual neurons optimize local PID objectives.

- **Mechanism:** A neuron  $N$  with inputs  $S_1, S_2$  might adjust its weights to maximize the **Unique Information** from  $S_1$  while minimizing **Redundancy**.
- **Result:** These networks self-organize into efficient processing architectures that mimic biological brains.
- **Geometric Link:** By minimizing redundancy, the network creates an "orthogonal" representation of the input space. In geometric terms, it is flattening the information manifold, attempting to approximate the "conformal flatness" that makes information



processing efficient.

## 6.2 Quantifying "Conscious" Processing

The distinction between Redundancy and Synergy is widely hypothesized to correlate with unconscious vs. conscious processing.

- **Redundancy (Robustness):** Unconscious, reflex actions. High redundancy ensures reliability (many paths to the same output).
- **Synergy (Integration):** Conscious experience. Requires the fusion of distinct sensory streams (visual + auditory) into a unified whole.
- **3-Way Hierarchy:** The 3-way analysis is crucial here. Consciousness is likely a higher-order phenomenon ( $n > 2$ ). Wibral's tools allow neuroscientists to track the "climb" up the lattice from low-level redundancy to high-level synergy as a stimulus enters awareness.

## 7. Conclusions and Key Takeaways

The investigation into the "exact fit" of the Lorentzian Lichnerowicz Conjecture and Wibral's 2025 PID work yields the following definitive conclusions:

1. **Structural Isomorphism of Rigidity:** The primary connection is **structural**. The Lichnerowicz Conjecture proves that **geometric causal structures** are rigid and resist conformal simplification. Wibral's Inconsistency Results prove that **information axiomatic structures** are rigid and resist decompositional simplification (additivity/invariance). Both fields have moved from seeking "flexible" measures/metrics to mapping the "hard constraints" of the system.
2. **Lorentzian Manifolds as the Causal Substrate:** The Lorentzian manifold provides the **physical definition of the sources** in PID. The light cone structure defines which variables *can* be redundant (spacelike separated, common past) and which *can* be synergistic (spacelike separated, common future). The Lichnerowicz conjecture guarantees that this causal definition is stable.
3. **The Breakdown of the Shannon Bit:** In the 3-way hierarchical lattice, the "Shannon bit" is revealed to be an aggregate of conflicting vector components. Wibral's work shows that we cannot define the "synergistic part of a bit" in a way that is universally consistent. The "bit" breaks down into incompatible atoms, just as "simultaneity" breaks down in relativity.
4. **Mereology as the Unified Language:** The adoption of mereology (lattice theory) in 2025 provides the common language. It describes both the **Causal Set** of the spacetime manifold and the **Redundancy Lattice** of the information decomposition.

In summary, the 2025 landscape reveals that information is not a fluid that fills a container; it is a geometric structure that must conform to the rigid causal topology of the Lorentzian universe. Wibral's inconsistency results are the information-theoretic equivalent of the existence of curvature—a proof that the simple, flat rules of the past (Shannon/Minkowski) are

only local approximations of a far more complex, interconnected reality.

## Appendix: Structured Comparisons

Table 1: The Wibral Inconsistency Trilemma (Dec 2025)

The following three properties cannot simultaneously hold for any PID measure.

| Property          | Definition                           | Informational Implication                      | Geometric Analogy                                  |
|-------------------|--------------------------------------|------------------------------------------------|----------------------------------------------------|
| Non-Negativity    | $\forall \alpha, \Pi(\alpha) \geq 0$ | "Knowledge" cannot be negative.                | Volume must be positive.                           |
| Target Chain Rule | $I(S; T_{\{1,2\}}) = \sum I(S; T_i)$ | $T_{\{i-1\}}$                                  | Information is additive across target splits.      |
| Invariance        | $\Pi(f(S)) = \Pi(S)$                 | Relabeling states does not change information. | Coordinate invariance / Diffeomorphism invariance. |

Table 2: The Redundancy Lattice vs. Lorentzian Causal Structure

| Structural Element | PID Redundancy Lattice (Mereology)  | Lorentzian Manifold (Causality) |
|--------------------|-------------------------------------|---------------------------------|
| Fundamental Unit   | Information Atom ( $\alpha$ )       | Event ( $p$ )                   |
| Ordering Relation  | Parthood ( $\alpha \preceq \beta$ ) | Causal Past ( $p \prec q$ )     |
| Intersection       | Shared Information (Redundancy)     | Past Light Cone Intersection    |
| Union              | Joint Information (Synergy + Red)   | Future Light Cone Union         |

|                       |                                        |                                                            |
|-----------------------|----------------------------------------|------------------------------------------------------------|
| <b>Constraint</b>     | Wibral Inconsistency<br>(Axioms clash) | Lichnerowicz Rigidity<br>(Symmetries clash with curvature) |
| <b>Metric/Measure</b> | Partial Information<br>$\pi(\alpha)$   | Spacetime Interval $ds^2$                                  |

**Table 3: Evolution of Concepts (2010 - 2025)**

| Concept   | Classical View<br>(Pre-2010) | Intermediate View<br>(2010-2024)   | Wibral 2025 View<br>(Current)                                                 |
|-----------|------------------------------|------------------------------------|-------------------------------------------------------------------------------|
| The Bit   | Scalar, Fungible unit.       | Decomposable into 4 atoms (2-way). | Tensor-like, axiom-dependent, structurally rigid.                             |
| Synergy   | Interaction term.            | "Extra" info in the whole.         | A lattice element that conflicts with the Chain Rule.                         |
| Causality | Correlation / Time-lag.      | Transfer Entropy.                  | Lorentzian geometric constraint on Lattice construction.                      |
| Goal      | Maximize Mutual Info.        | Maximize Synergy/Redundancy.       | Optimize Local Goal Functions (Infomorphic Networks) within axiomatic limits. |

**Works cited**

1. (PDF) Novel Inconsistency Results for Partial Information Decomposition - ResearchGate, accessed January 1, 2026, [https://www.researchgate.net/publication/398850672\\_Novel\\_Inconsistency\\_Results\\_for\\_Partial\\_Information\\_Decomposition](https://www.researchgate.net/publication/398850672_Novel_Inconsistency_Results_for_Partial_Information_Decomposition)
2. Novel Inconsistency Results for Partial Information Decomposition - arXiv, accessed January 1, 2026, <https://arxiv.org/html/2512.16662v1>

3. DECOMPOSITION THEOREMS FOR LORENTZIAN MANIFOLDS WITH NONPOSITIVE CURVATURE - Project Euclid, accessed January 1, 2026, <https://projecteuclid.org/journals/journal-of-differential-geometry/volume-22/issue-1/Decomposition-theorems-for-Lorentzian-manifolds-with-nonpositive-curvature/10.4310/jdg/1214439719.pdf>
4. Gravity and Matter in Causal Set Theory - arXiv, accessed January 1, 2026, <https://arxiv.org/pdf/0801.0240>
5. accessed January 1, 2026, [https://www.researchgate.net/publication/373710054\\_The\\_Lorentzian\\_Lichnerowicz\\_conjecture\\_for\\_real-analytic\\_three-dimensional\\_manifolds#:~:text=In%20the%20second%20part%2C%20we,manifold%20must%20be%20conformally%20flat](https://www.researchgate.net/publication/373710054_The_Lorentzian_Lichnerowicz_conjecture_for_real-analytic_three-dimensional_manifolds#:~:text=In%20the%20second%20part%2C%20we,manifold%20must%20be%20conformally%20flat)
6. A local Lorentzian Ferrand-Obata theorem for conformal vector fields - arXiv, accessed January 1, 2026, <https://arxiv.org/html/2511.03713v1>
7. The conformal group of a - AMS :: Journal of the American Mathematical Society, accessed January 1, 2026, <https://www.ams.org/jams/2022-35-01/S0894-0347-2021-00976-1/>
8. 1 Introduction - arXiv, accessed January 1, 2026, <https://arxiv.org/html/2309.11184v2>
9. Novel Inconsistency Results for Partial Information Decomposition - arXiv, accessed January 1, 2026, <https://arxiv.org/pdf/2512.16662>
10. A systematic approach to brain dynamics: cognitive evolution theory of consciousness, accessed January 1, 2026, [https://www.researchgate.net/publication/362608424\\_A\\_systematic\\_approach\\_to\\_brain\\_dynamics\\_cognitive\\_evolution\\_theory\\_of\\_consciousness](https://www.researchgate.net/publication/362608424_A_systematic_approach_to_brain_dynamics_cognitive_evolution_theory_of_consciousness)
11. The stream of consciousness. a In brain dynamics, every conscious state... - ResearchGate, accessed January 1, 2026, [https://www.researchgate.net/figure/The-stream-of-consciousness-a-In-brain-dynamics-every-conscious-state-evolves-from-the\\_fig2\\_362608424](https://www.researchgate.net/figure/The-stream-of-consciousness-a-In-brain-dynamics-every-conscious-state-evolves-from-the_fig2_362608424)
12. Neural timescales from a computational perspective - arXiv, accessed January 1, 2026, <https://arxiv.org/html/2409.02684v2>
13. Synchronisation through Integrated Information in Neuropercolation Models near Criticality - repository, ULisboa, accessed January 1, 2026, [https://repositorio.ulisboa.pt/bitstream/10451/63924/1/TM\\_Timofej\\_Luitle.pdf](https://repositorio.ulisboa.pt/bitstream/10451/63924/1/TM_Timofej_Luitle.pdf)
14. (a) Information diagram depicting the information provided by statement... - ResearchGate, accessed January 1, 2026, [https://www.researchgate.net/figure/a-Information-diagram-depicting-the-information-provided-by-statement-A-and-C-If\\_fig3\\_353085162](https://www.researchgate.net/figure/a-Information-diagram-depicting-the-information-provided-by-statement-A-and-C-If_fig3_353085162)
15. Aaron J. Gutknecht - Google Scholar, accessed January 1, 2026, <https://scholar.google.com/citations?user=3luSqaYAAAAJ&hl=de>
16. 248336 PDFs | Review articles in DECOMPOSITION - ResearchGate, accessed January 1, 2026, <https://www.researchgate.net/topic/Decomposition/publications>
17. [2512.16662] Novel Inconsistency Results for Partial Information Decomposition - arXiv, accessed January 1, 2026, <https://arxiv.org/abs/2512.16662>
18. A general framework for interpretable neural learning based on local

information-theoretic goal functions - NIH, accessed January 1, 2026,  
<https://pmc.ncbi.nlm.nih.gov/articles/PMC11912414/>

19. A general framework for interpretable neural learning based on local information-theoretic goal functions | Request PDF - ResearchGate, accessed January 1, 2026,  
[https://www.researchgate.net/publication/389598572\\_A\\_general\\_framework\\_for\\_interpretable\\_neural\\_learning\\_based\\_on\\_local\\_information-theoretic\\_goal\\_functions](https://www.researchgate.net/publication/389598572_A_general_framework_for_interpretable_neural_learning_based_on_local_information-theoretic_goal_functions)